Smart Campus Equipment & Lab Inventory Management System

Product Name:

Team Members:

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Project Description:

Colleges often face difficulties managing lab equipment due to manual record-keeping, making it hard to track available items, borrowed equipment, and who has taken what. Our Project aims to provide a centralized platform where students and faculty can check equipment availability, view borrowing status, and track who has borrowed specific items, making inventory management simpler, faster, and more transparent.

Project Phases:

Phase 1: Project Validation and Requirements

Validate the problem we are solving, understand the needs of students and lab staff, and finalize the project's functional and non-functional requirements.

Phase 2: Backend Research and Data Collection

Collect the necessary information about lab equipment, inventory records, borrowing procedures, users, and other details required to design the backend and database.

Phase 3: Application Development and Testing

Build the application, implement features such as equipment availability tracking, borrowing records, and test the system to identify and fix issues.

Phase 4: Deployment and User Implementation

Deploy the application and make it available for students and lab staff to use. Gather user feedback, identify areas for improvement, and refine the application based on real-world usage.

Project Timeline:

SL NO	Start Date	End Date
Phase 1		
Phase 2		
Phase 3		
Phase 4		

Product Manual:

PHASE 1

Project Validation:

We interacted with laboratory instructors, faculty, and students to understand the existing challenges in managing lab equipment and inventory. Based on their inputs, we identified the need for a centralized platform to check equipment availability and track borrowing records, reducing dependency on manual processes.









Stake Holders:

Students, Lab Instructors, Faculty, Researchers, University Management

Actor	Description
Student / Researchers	Searches equipment, requests items, views own borrowing history
Faculty	Same as student, plus reserves equipment for lab sessions
Lab Assistant / Technician	Approves requests, records issue and return, reports damage
Admin (Lab In-charge / HOD)	Manages inventory, users, maintenance and reports

Functional Requirements

Inventory Management

ID	Requirement
FR1	Admin/Lab Assistant can add, edit and delete equipment (name, category, serial no., lab, quantity, condition, purchase date).
FR2	Each item has a unique ID with a QR code/barcode for quick scanning.
FR3	System shows real-time **Status** (Available, Reserved, Issued, Under Maintenance, Lost) and a separate **Condition** (Good, Minor Damage, Damaged, Unusable) for every item.
FR4	Search and filter by name, category, lab or status.

Issue / Return Workflow

ID	Requirement
FR5	Students/faculty can request equipment online; the lab assistant approves or rejects.
FR6	*Lab assistant records issue and return; due date is set automatically.*
FR7	System blocks issuing items that are unavailable, reserved by another user, or under maintenance.
FR8	Return logs the item's condition; damaged items are flagged and can be moved to Under Maintenance.

Notifications and Tracking and Maintenance

ID	Requirement
FR9	Reminders before due date and overdue alerts to borrower and lab assistant
FR10	low stock alerts for consumables e.g. resistors chemicals
FR11	full audit trial of who borrowed what and when
FR12	users can report faulty equipment, admin schedules and tracks repairs

Users and reports

ID	Requirements
FR13	Registration-login with role-based access control
FR14	reports for usage, overdue items, damaged items and inventory summary, exportable as PDF or CSV.

Reservations and Request Management

ID	Requirement
FR15	Students/faculty can reserve available equipment for a future date and time. The system shall prevent conflicting reservations for the same equipment. A reservation not collected within 30 minutes of its slot start expires automatically.
FR16	A user can cancel a pending request or reservation (including one already approved) before the equipment is issued; the equipment then returns to Available.
FR17	Equipment quantity and available stock cannot become negative; a request exceeding available stock is rejected or limited to the available quantity.

Non-Functional Requirements

Category	Requirement
Performance	Common operations (search, issue, return) respond within 2 seconds with up to 50 concurrent users.
Security	Hashed passwords, role-based permissions, session timeout.
Usability	Works on mobile and desktop; simple enough for a new user without training.
Reliability	99% uptime and daily data backup.
Scalability	Supports multiple labs and departments.
Data Integrity	No two users can be issued the same item at once.

Future Implementation Plan

- Penalty or restriction for repeated late returns
- Dashboard charts showing most-used and most-damaged equipment

Test Cases

ID	Req.	Scenario	Input / Steps	Expected Result
TC1	FR13	Valid login	Enter correct credentials	Redirected to role-based dashboard
TC2	FR13	Invalid login	Enter wrong password	Error message; access denied
TC3	FR13	Role restriction	Student opens admin page	Access denied
TC4	FR1	Add equipment	Fill all valid fields	Item saved and appears in inventory
TC5	FR1	Add with missing fields	Leave serial no. blank	Validation error shown
TC6	FR1	Duplicate serial no.	Add an existing serial no.	Rejected with "already exists"
TC7	FR2	QR scan	Scan item QR code	Opens that item's details
TC8	FR4	Search	Search "oscilloscope"	Matching items listed
TC9	FR4	Search, no match	Search "xyz123"	"No results found" message
TC10	FR5	Request item	Student requests available item	Request goes to lab assistant as Pending
TC11	FR6	Issue item	Assistant approves request	Status becomes Issued; due date set
TC12	FR7	Issue unavailable item	Request an already issued item	System blocks the request
TC13	FR7	Concurrent request	Two users request the last unit at once	Only one succeeds
TC14	FR8	Normal return	Return item in good condition	Status returns to Available
TC15	FR3, FR8	Damaged return	Set condition to Damaged on return	Condition = Damaged; status set to Under Maintenance; admin notified
TC16	FR9	Due reminder	1 day before due date	Reminder sent to borrower
TC17	FR9	Overdue alert	Due date passes without return	Overdue flag set; alert sent
TC18	FR10	Low stock	Consumable falls below threshold	Alert sent to admin
TC19	FR11	History	Open an item's history	Shows all past issues/returns in order
TC20	FR12	Report fault	User submits fault report	Ticket created; status updated
TC21	FR14	Export report	Export overdue report as CSV	File downloads with correct data
TC22	NFR	Load test	50 users search at once	Response under 2 seconds
TC23	NFR	Session timeout	Stay idle for 15 minutes	Auto logout
TC24	NFR	Mobile view	Open on a phone	Layout adapts; all features usable
TC25	FR15	Valid reservation	Reserve available equipment for a future slot	Reservation confirmed; status shows Reserved for that slot
TC26	FR15	Conflicting reservation	Another user reserves the same equipment for the same period	System rejects the conflicting reservation
TC27	FR16	Cancel request	Student cancels a pending request	Request status becomes Cancelled; equipment remains Available
TC28	FR17	Quantity exceeds stock	3 Arduino boards available; user requests 5	Request rejected or limited to 3; stock never goes negative

PHASE 2